

Truthmaker Semantical Approaches to Belief Revision

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Abstract

In this paper, I propose some frameworks for Belief Revision by using resources from Finean truthmaker semantics, resembling the traditional AGM approach. New frameworks will be able to determine a unique contraction, and typically lack some disputed features such as logical omniscience and recovery.

1 Introduction

In this paper, I will develop a family of hyperintensional frameworks for belief revision utilizing the resources provided by Kit Fine’s truthmaker semantics.

The classical AGM framework (notably Alchourrón, Gärdenfors, and Makinson, 1985; the description below follows that of van Ditmarsch, van der Hoek, and Kooi, 2008) represents an agent’s doxastic state as a theory or belief set $\mathcal{K}$ in a propositional language. Let $\mathcal{L}$ be the propositional language generated by a countable set of atomic propositions P under the classical truth-functional connectives. Let Cn be the classical consequence operator, where $\text{Cn}(\Sigma) = \{\phi \in \mathcal{L} \mid \Sigma \vdash \phi\}$. A *belief set* $\mathcal{K} \subseteq \mathcal{L}$ is defined as a set of formulas closed under Cn , i.e., $\mathcal{K} = \text{Cn}(\mathcal{K})$. The unique inconsistent belief set is denoted by $\mathcal{K}_\perp = \mathcal{L}$.

The AGM framework includes three basic operations that map a belief set $\mathcal{K}$ and an input sentence $\phi \in \mathcal{L}$ to a new belief set. Intuitively,

1. **Expansion** ($\mathcal{K} \oplus \phi$): The incorporation of new information ϕ into $\mathcal{K}$, without retracting

any existing beliefs.

2. **Contraction** ($\mathcal{K} \ominus \phi$): The removal of a belief ϕ from $\mathcal{K}$, such that the resulting set no longer implies ϕ , unless ϕ is a classical tautology ($\vdash \phi$).
3. **Revision** ($\mathcal{K} \circledast \phi$): The integration of a new, potentially conflicting belief ϕ into $\mathcal{K}$ (if ϕ is consistent), while possibly some sentences in $\mathcal{K}$ should be removed so the resulting set is also consistent.

The specific behavior of expansion is characterized by the postulates $(\oplus 1)$ - $(\oplus 6)$ in Table

1. This additionally captures the idea that $\mathcal{K} \oplus \phi = \text{Cn}(K \cup \{\phi\})$.

The postulates for expansion determine a clear and unique result after expansion, whereas characterizing contraction is more complicated under the AGM approach. Here I will not go deeper beyond so-called *partial meet contraction*, which conforms to the postulates $(\ominus 1)$ - $(\ominus 6)$ in 2. The idea behind it is roughly that if ϕ is contracted from $\mathcal{K}$, the resulting belief set is the intersection of the favored maximal subsets of $\mathcal{K}$ that does not imply ϕ . If ϕ is not in $\mathcal{K}$, then the maximal subset of $\mathcal{K}$ that does not imply ϕ would be $\mathcal{K}$ itself.

$(\oplus 1)$ $\mathcal{K} \oplus \phi$ is a belief set	<i>type</i>
$(\oplus 2)$ $\phi \in \mathcal{K} \oplus \phi$	<i>success</i>
$(\oplus 3)$ $\mathcal{K} \subseteq \mathcal{K} \oplus \phi$	<i>expansion</i>
$(\oplus 4)$ $\phi \in \mathcal{K} \implies \mathcal{K} = \mathcal{K} \oplus \phi$	<i>minimal action</i>
$(\oplus 5)$ For all $\mathcal{H}, \mathcal{K} \subseteq \mathcal{H} \implies \mathcal{K} \oplus \phi \subseteq \mathcal{H} \oplus \phi$	<i>monotony</i>
$(\oplus 6)$ $\mathcal{K} \oplus \phi$ is the smallest set satisfying $(\oplus 1) - (\oplus 5)$	<i>minimal change</i>

Table 1: The AGM Postulates for Expansion.

$(\ominus 1)$ $\mathcal{K} \ominus \phi$ is a belief set	<i>type</i>
$(\ominus 2)$ $\mathcal{K} \ominus \phi \subseteq \mathcal{K}$	<i>contraction</i>
$(\ominus 3)$ $\phi \notin \mathcal{K} \implies \mathcal{K} = \mathcal{K} \ominus \phi$	<i>minimal action</i>
$(\ominus 4)$ $\not\vdash \phi \implies \phi \notin \mathcal{K} \ominus \phi$	<i>success</i>
$(\ominus 5)$ $\phi \in \mathcal{K} \implies \mathcal{K} \subseteq (\mathcal{K} \ominus \phi) \oplus \phi$	<i>recovery</i>
$(\ominus 6)$ $\vdash \phi \leftrightarrow \psi \implies \mathcal{K} \ominus \phi = \mathcal{K} \ominus \psi$	<i>extensionality</i>

Table 2: The Basic AGM Postulates for Contraction.

Once expansion and contraction are characterized, we can define revision in terms of these two operations by so-called *Levi Identity*:

$$\text{Levi Identity: } K \circledast \phi = (K \ominus \neg\phi) \oplus \phi$$

The revision operation, defined with the expansion and the partial meet contraction conforming to the postulates above can be axiomatized via the postulates $(\circledast 1)$ - $(\circledast 6)$ in Table 3.

$(\circledast 1)$ $\mathcal{K} \circledast \phi$ is a belief set	<i>type</i>
$(\circledast 2)$ $\phi \in \mathcal{K} \circledast \phi$	<i>success</i>
$(\circledast 3)$ $\mathcal{K} \circledast \phi \subseteq \mathcal{K} \oplus \phi$	<i>upper bound</i>
$(\circledast 4)$ $\neg\phi \notin \mathcal{K} \implies \mathcal{K} \oplus \phi \subseteq \mathcal{K} \circledast \phi$	<i>lower bound</i>
$(\circledast 5)$ $\mathcal{K} \circledast \phi = \mathcal{K}_\perp \iff \vdash \neg\phi$	<i>triviality</i>
$(\circledast 6)$ $\vdash \phi \leftrightarrow \psi \implies \mathcal{K} \circledast \phi = \mathcal{K} \circledast \psi$	<i>extensionality</i>

Table 3: The AGM Postulates for Revision.

It would be redundant to spill more ink to praise the exquisiteness and the usefulness of the AGM approach, since it already has been appreciated for several decades. Yet for some reason one might desire different kinds of frameworks. In section 1.1, I will propose the desiderata of the framework I will develop in this paper. Section 1.2 will be devoted to deploy the conceptual materials from Finean truthmaker semantics. Section 2 presents two simple frameworks that partially satisfies the desiderata in section 1.1. Several slightly more sophisticated frameworks which is stronger than those presented in §2 is developed in section 3, which fully satisfies the desiderata with different degrees and means. Finally, the paper will be concluded with a formal appendix.

1.1 Desiderata

1.1.1 Definite Contraction

The AGM framework assumes unique contraction. That is, by $(\ominus 1)$, the result of contracting ϕ from $\mathcal{K}$ must be one particular belief set. But the postulates $(\ominus 1)$ - $(\ominus 6)$ by themselves do not guarantee there is a unique belief set that satisfies them.

For example, suppose $\mathcal{K} = \text{Cn}(\{p, q, r\})$. In this case $\mathcal{K} \ominus (p \wedge q)$ has multiple possible outcomes. By $(\ominus 4)$ and the definition of belief set, it would be evident that at least one of p and q should be removed from the belief set; but which one? Either choice satisfies the postulates, and even if one removes both of them, it is still a valid partial meet contraction.

But then, how do we choose one among them? In partial meet contraction, the choice was somewhat arbitrary.

The standard way out for the arbitrariness was by incorporating two more postulates ($\ominus 7$) and ($\ominus 8$) - which results in *transitively relational partial meet contraction*. This turned out to capture the idea of *epistemic entrenchment* (Gärdenfors & Makinson, 1988). In case where the epistemic agent is more entrenched into p , $\mathcal{K} \ominus (p \wedge q)$ would be $\text{Cn}(p, r)$, whereas in case it is more entrenched into q , the result would be $\text{Cn}(q, r)$.

$(\ominus 7) ((\mathcal{K} \ominus \phi) \cap (\mathcal{K} \ominus \psi)) \subseteq \mathcal{K} \ominus (\phi \wedge \psi)$	<i>min-conjunction</i>
$(\ominus 8) \phi \notin \mathcal{K} \ominus (\phi \wedge \psi) \implies \mathcal{K} \ominus (\phi \wedge \psi) \subseteq \mathcal{K} \ominus \phi$	<i>max-conjunction</i>

Table 4: Additional AGM Postulates for Contraction.

Yet one could still complain that the entrenchment cannot be inferred from the belief set alone. The transitively relational partial meet contraction forces a regularity on contracting some formulas from $\mathcal{K}$, but still does not by itself decide what the regularity is. The regularity can be determined only if the entrenchment is already specified. But to fully specify the denumerably long list of entrenchment is, for one thing, practically not very realistic. Thus, one might desire a framework which can determine the result of contraction simply from what is contracted from what, just as expansion in the AGM framework is definite in that sense.

A related consideration is that one might want to model the epistemic entrenchment as a subject of change as the belief set undergoes some operations. Let $\mathcal{K} = \text{Cn}(\{p, \neg q\})$. It seems natural to me to suppose that if $\mathcal{K}$ is revised with $p \wedge q$, not only is $\neg q$ contracted, but also that p is possibly in some sense *reinforced*, so p might be more entrenched to the agent after the operation of revision. Thus, one might desire entrenchment to be understood in terms of operations such as expansion and contraction; not to determine the result of contraction via entrenchment.

So this gives the first desideratum for frameworks I will develop in this paper:

(D1) **Definite Contraction:** *The definition, or the postulates of the contraction operator alone should be sufficient to determine the result after the contraction*

1.1.2 Logical Ignorance, Paraconsistency, and Local Rationality

Logical omniscience is a traditional worry for any kinds of epistemic logic with the necessitation rule: $\vdash \phi \implies \vdash \Box \phi$. In epistemic logic, this gives that an epistemic agent has the doxastic or epistemic attitude which is supposed to conform to the behavior of $\Box$, towards all logical truths. Usually this yields the consequence that an ideal epistemic agent must know or believe all logical truths, which may seem suspicious.

The AGM approach yields the same result. Since the belief set $\mathcal{K}$ is defined as a set of formulas closed under the classical consequence operator, in any case the epistemic agents believes all the logical truths, which causes the same discomfort as the problem of logical omniscience for epistemic logics.

Additional worry is that the logical omniscience results in dismissing the contraction with logical truths. This is what *minimal action* postulate commands; but even without the postulate the definition of $\mathcal{K}$ itself forbids the removal of any logical truths. Revising *minimal action* with a weaker postulate at most only results in eccentric possibilities that by contracting a logical truth it rather takes out some innocent formulas other than what is contracted. However, one might want to model what an ideally rational epistemic agent must believe after it is forced to contract a logical truth from the belief set.

Likewise, one might feel uncomfortable to the idea that one must believe all the logical consequences of its belief in general. One case we might want to model is the pessimism for one's own belief. One might be equally entrenched deeply to each of the 'atomic' belief, but came to contract the conjunction of all of its beliefs, for it also has certainty towards at least some of its belief would be wrong; that the agent itself cannot be perfect. In this case, a faithful modelling seems not to remove all the beliefs (or all the logically intermediate beliefs) from the original belief set, but only to remove the conjunction, yet retaining all the 'atomic' belief. But to this one must admit a belief set that is not closed under classical consequence.

So this gives the second desideratum:

(D2) **Logical Ignorance:** *The belief set must be able to exclude its logical consequences*

There is also a similar worry for expansion. Again, since $\mathcal{K}$ is closed under the classical consequence, starting with inconsistent beliefs, or expanding with any ϕ such that $\mathcal{K} \vdash \neg\phi$ results in explosion.

But it may be unclear why an ideally rational epistemic agent should be so vulnerable to inconsistency. Especially, one might want to model how should one operate when the initial state of belief set is reasonably inconsistent. This gives the third desideratum:

(D3) **Paraconsistency**: *The belief set must be able to admit inconsistency without explosion*

Finally, the motivating idea behind **Logical Ignorance** and **Paraconsistency** may be the consideration that we might want to model an epistemic agent that is *locally* rational, in the sense that it contemplates a subset of the whole belief set at a time. Indeed, human epistemic agents never inspect every corner of our own belief set when expanding or contracting our beliefs; and this is somewhat desirable even for an ideal epistemic agent when one considers cognitive economy.

This is also motivated from the idea that the strength of those two desiderata must be somehow inhibited. For example, **Paraconsistency**, in its extreme, would allow one to believe apparent contradictions such as $\phi \wedge \neg\phi$; yet we might want the modelled ideal epistemic agent be aware enough not to make such a silly mistake. **Logical Ignorance**, on the other hand, would make possible that the ideal agent believes both ϕ and $\neg\phi \vee \psi$ ($\iff \phi \rightarrow \psi$) without believing ψ . Although we might be reluctant to take the modelled agent to be perfect in all logical regards, we still want it to be locally *rational*.

So this gives the fourth, somewhat vague desideratum:

(D4) **Local Rationality**: *The modelled epistemic agent must be rational, but only locally so.*

1.1.3 Hyperintensionality

In §1.1.2, it was suggested that the new framework must be able to contract with logical truths. Naturally, if it is possible to contract with logical truths, it would seem also possible to expand, or revise with logical truths.

Yet with *extensionality* postulates for contraction and revision, non-trivial contraction and revision with logical truths consist only of two kinds: either you contract *all* logical truths, or you revise and thus incorporate *all* logical truths. Suppose the belief set $\mathcal{K}$ such that $\{p \vee \neg p, q \vee \neg q\} \subset \mathcal{K}$ and $r \vee \neg r \notin \mathcal{K}$. This would be possible in frameworks where **Logical Ignorance** is satisfied. Now consider $\mathcal{K} \ominus (p \vee \neg p)$. Since $\vdash (p \vee \neg p) \leftrightarrow (q \vee \neg q)$, by *extensionality* of contraction $\mathcal{K} \ominus (p \vee \neg p) = \mathcal{K} \ominus (q \vee \neg q)$, and by *success* we get $\{p \vee \neg p, q \vee \neg q\} \cap \mathcal{K} \ominus (p \vee \neg p) = \emptyset$. Furthermore, by extensionality for revision, likewise we get $\{p \vee \neg p, q \vee \neg q, r \vee \neg r\} \subset (\mathcal{K} \ominus (p \vee \neg p)) \circledast (r \vee \neg r)$.

But one might desire a more fine-grained contraction and revision. For one thing, if one already has **Local Rationality**, one might want to be able to contract some particular kind of logically true formulas without affecting other formulas, especially for those have different *subject matter*. Likewise, one might want to be able to expand, or revise with some logically true formulas without bringing infinities of logical truths into the belief set.

So this gives the fifth desideratum:

- (D5) **Hyperintensionality**: Two extensionally, or intensionally equivalent but hyperintensionally distinct formulas must be able to behave differently upon expansion or contraction.

1.1.4 Pertinent Recovery

Finally, it should be noted that the *recovery* postulates was a subject of dispute for a long time. The idea that if one loses a belief and immediately regains it the agent must at least return to the original belief set would seem intuitively compelling. The postulate also plays a crucial role in preventing the contraction from being unnecessarily strong, so contracting with ϕ cannot remove more formulas from the belief set than what expanding with ϕ would take into the belief set.

Yet *recovery*, as the AGM framework postulates, also yields counterintuitive results. Suppose $\{p, q, p \wedge q, p \vee q\} \subset \mathcal{K}$. Now, the fact that $\{p, q, p \wedge q, p \vee q\} \cap (\mathcal{K} \ominus (p \vee q)) = \emptyset$ will follow in the AGM framework (by *success* of contraction and the definition $\mathcal{K} = \text{Cn}(\mathcal{K})$), and it is quite intuitive. If the modelled epistemic agent loses the belief that it is either the case that p or q ,

reasonably the agent would also lose the belief that it is both the case that p and q . So far so good, but *recovery* yields that if the modelled agent immediately regains the belief that $p \vee q$, the agent must also regain the belief that $p \wedge q$. That is: $\{p, q, p \wedge q, p \vee q\} \subset (\mathcal{K} \ominus (p \vee q)) \oplus (p \vee q)$. But this is intuitively not very compelling.

To generalize: suppose $\{\phi, \psi\} \subset \mathcal{K}$ where ϕ strictly entails ψ but not the other way around. This yields the result that $\phi \notin \mathcal{K} \ominus \psi$. Yet by recovering ψ , the modelled agent recovers not only ψ but the stronger ϕ . This is also suspicious, for recovering ψ seems to have no reason to involve ϕ if ψ does not entail ϕ , regardless of the fact that ϕ was in the initial belief set before the contraction (Hansson, 2022).

It seems to me that the intuition behind proposed counterexamples for *recovery* is that one might have different *sources* for the same belief. If ψ is to be contracted from $\mathcal{K}$, not only should the *sources*, or reason for believing ψ should be discarded, but also should the sources of the belief that ϕ be defeated, for any source of the belief that ϕ would also be the source of the belief that ψ . But when one recovers the belief that ψ , it is not guaranteed that the recovery is via reconfirming the exact source for both beliefs that ψ and ϕ . One may recover the belief that ψ via *new* source for the belief that ψ which does not make one believe that ϕ . It is possible because, since ψ does not entail ϕ , the source for the belief that ψ is not necessarily the source for the belief that ϕ .

Notice how *recovery* is extremely compelling when it is formulated in terms of the source of belief: if a source of belief is once discarded and immediately reconfirmed, the original belief set must be a subset of the belief set after the sequence. Thus, one might wish the framework to be able to grasp this sense of recovery, not the one among the AGM postulates.

This gives the final, sixth desideratum:

(D6) **Pertinent Recovery** The framework must avoid the problematic case of *recovery*.

1.2 Materials From Truthmaker Semantics

In this paper, I propose frameworks that satisfy **Definite Contraction**, **Logical Ignorance**, **Paraconsistency**, **Hyperintensionality**, and **Pertinent Recovery**. **Local Rationality** will not be satisfied in two prototypes, and will be realized in other frameworks in different degrees.

The core idea is to posit a *belief-state set* $\mathcal{S}^{\mathcal{K}}$, where any belief-state set determines a unique belief set. Intuitively, $\mathcal{S}^{\mathcal{K}}$ is the set of the *source* of the modelled agent's belief; so any member of $\mathcal{S}^{\mathcal{K}}$ is a state which *verifies*, or *truthmakes* some member of $\mathcal{K}$. Operations are defined to operate on $\mathcal{S}^{\mathcal{K}}$ instead of $\mathcal{K}$; thus we expand, contract or revise with a state σ from a belief-state set $\mathcal{S}^{\mathcal{K}}$. The expansion, contraction or revision with a formula ϕ from a belief set $\mathcal{K}$ is only defined in terms of the operation with state on a belief-state set.

The ideas presented in this subsection originates from Kit Fine's work on truthmaker semantics. Below in this subsection I introduce some important resources from Finean truthmaker semantics which will be directly relevant to new frameworks developed in this paper. The following details are mostly taken from Fine (2016), Fine (2017a), Fine (2017b), and Fine (2017c).

A state, informally, is anything that exactly verifies or exactly falsifies a statement (the notion of exact and inexact verification/falsification will be covered shortly). We do not need to assume a particular ontology of states to do semantics, as we do not need to assume a particular ontology of possible worlds to do modal logics.

One characteristic feature of states is that states can have mereological structure. Thus a state can be a part of another state; and any two states can fuse to compose a single state. The parthood relation here is supposed to be antisymmetric and transitive.

Formally, we achieve such an understanding on states by defining a *set space*. To do so we first define the parthood relation $\sqsubseteq$ and the notion of *least upper bound*.

Definition 1.1 (Parthood Relation on S). $\sqsubseteq$ (part) is a binary relation on a set of states S which conforms to the following three conditions:

$$\text{Reflexivity: } \forall s (s \sqsubseteq s)$$

Antisymmetry: $\forall s \forall t ((s \sqsubseteq t \& t \sqsubseteq s) \implies s = t)$

Transitivity: $\forall s \forall t \forall u ((s \sqsubseteq t \& t \sqsubseteq u) \implies s \sqsubseteq u)$

Definition 1.2 (Least Upper Bound). *A state s is an upper bound of $T \subseteq S$ iff $\forall t \in T \ t \sqsubseteq s$.*

A state s is a least upper bound of T iff s is an upper bound of T , and it is part of any upper bound s' of T .

Definition 1.3 (State Space). *A state space is an ordered set $\langle S, \sqsubseteq \rangle$, where S is a set of states and $\sqsubseteq$ (part) is a binary relation on S , and for any $T \subseteq S$ its least upper bound exists.*

It will be easy to check that the least upper bound is unique by antisymmetry. The least upper bound (sometimes called ‘fusion’) of T is denoted by $\sqcup T$, or $t_1 \sqcup t_2 \sqcup t_3 \dots$ for $T = \{t_1, t_2, t_3, \dots\}$. Also, it would be easy to check that if $s' \sqsubseteq s$, $s' \sqcup s = s$, since s and s' are both part of s and all upper bounds of $\{s', s\}$ must by definition have s as part.

Two extreme states are worth noting:

Convention 1.1 (Null State and Full State). *Let ‘ $\square$ ’ denote the least upper bound of $\emptyset \subseteq S$, which is a part of any state and has no part. This will also be called ‘null state’. Let ‘ $\blacksquare$ ’ denote the least upper bound of $S \subseteq S$, which is not a part of any state and has every state as part. This will also be called ‘full state’*

The definition for the propositional language we will use throughout the paper is defined as follows:

Definition 1.4 (Propositional Language $\mathcal{L}$). *Let P be the set of atomic formulas. The language $\mathcal{L}$ is generated by the following BNF:*

$$\phi ::= p \mid \neg \phi \mid (\phi \wedge \phi) \mid (\phi \vee \phi)$$

for any $p \in P$.

We also deploy standard conventions for bracketing and definitions for other boolean connectives. Notably, $(\phi \rightarrow \psi)$ is defined as $(\neg \phi \vee \psi)$.

The *truthmaker model*, intuitively, decides a pair of sets - the set of verifiers and the set of falsifiers - of states in a state space for each atomic formulas. Given a model we can recursively

define the verification and falsification condition for any formula in $\mathcal{L}$. Formally:

Definition 1.5 (Modalized Set Space). *A modalized set space is a triple $\langle S, S^\diamond, \sqsubseteq \rangle$ where $\langle S, \sqsubseteq \rangle$ is a set space and $S^\diamond$ (set of all possible states) is a non-empty subset of S such that $(s \in S^\diamond \ \& \ t \sqsubseteq s) \implies t \in S^\diamond$.*

Definition 1.6 (Model M). *A model M is a quadruple $\langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$ where $\langle S, S^\diamond, \sqsubseteq \rangle$ is a modalized set space, and $|\cdot|$ is a valuation function which takes every $p \in P$ to a pair $\langle V, F \rangle$, where V and F are non-empty subsets of S that conforms to the following conditions:*

$$\text{Exclusivity: } \forall v \in V \ \forall f \in F \ v \sqcup f \notin S^\diamond$$

$$\text{Exhaustivity: } \forall s \in S^\diamond (\exists v \in V \ s \sqcup v \in S^\diamond \vee \exists f \in F \ s \sqcup f \in S^\diamond)$$

Given $|p| = \langle V, F \rangle$, let $|p|^+$ denote the first component (i.e. V), and $|p|^-$ denote the second component (i.e. F).

Alternatively, we could allow V or F to be empty but never both. In fact, *Exhaustivity* already implies that it can never be the case that both V and F are empty, and *Exclusivity* grants that if one of them is empty, the other one would be $\{\square\}$.

Given a model M , we define a basic notion of verification and falsification, that is, *exact* verification $\Vdash$ and *exact* falsification $\dashv\vdash$:

Definition 1.7 (Exact Verification and Falsification for Formulas). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$,*

$$(atom)^+ \ s \Vdash p \iff s \in |p|^+$$

$$(atom)^- \ s \dashv\vdash p \iff s \in |p|^-$$

$$(negation)^+ \ s \Vdash \neg\phi \iff s \dashv\vdash \phi$$

$$(negation)^- \ s \dashv\vdash \neg\phi \iff s \Vdash \phi$$

$$(conjunction)^+ \ s \Vdash (\phi \wedge \psi) \iff \exists t, u \text{ s.t. } t \Vdash \phi \ \& \ u \Vdash \psi \ \& \ s = t \sqcup u$$

$$(conjunction)^- \ s \dashv\vdash (\phi \wedge \psi) \iff s \dashv\vdash \phi \text{ or } s \dashv\vdash \psi$$

$$(disjunction)^+ \ s \Vdash (\phi \vee \psi) \iff s \Vdash \phi \text{ or } s \Vdash \psi$$

$$(disjunction)^- \quad s \dashv\!\parallel (\phi \vee \psi) \iff \exists t, u \text{ s.t. } t \dashv\!\parallel \phi \ \& \ u \dashv\!\parallel \psi \ \& \ s = t \sqcup u$$

Notice that the exact verifier of $\phi \wedge \psi$ does not always exactly verifies ϕ . Likewise, the exact falsifier of $\phi \vee \psi$ does not always exactly falsifies ϕ . Intuitively, a state s exactly verifies ϕ only if s is wholly relevant with the truth of ϕ . For example, the state that it is raining is wholly relevant with the truth of the statement “it is raining”, but the state that it is raining and there is a dog lying in the gutter is not wholly relevant with the truth of the statement “it is raining”; it is merely *partially* relevant. Similar with the case of exact falsification.

Yet we could say the state that it is raining and there is a dog lying in the gutter *inexactly* verifies the statement “it is raining”. With the mereological structure of states in hand, we can define *inexact verification* $\Vdash$ and *inexact falsification* $\dashv\!\parallel$ as follows:

Definition 1.8 (Inexact Verification and Falsification for Formulas). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, | \cdot | \rangle$,*

$$(inexact)^+ \quad s \Vdash \phi \iff \exists t \sqsubseteq s \text{ s.t. } t \Vdash \phi$$

$$(inexact)^- \quad s \dashv\!\parallel \phi \iff \exists t \sqsubseteq s \text{ s.t. } t \dashv\!\parallel \phi$$

Finally, we define the notion of *subject matter*:

Definition 1.9 (Subject Matter). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, | \cdot | \rangle$, let $\mathcal{V}(\phi)$ denote the set of all exact verifiers of ϕ ; and $\mathcal{F}(\phi)$ denote the set of all exact falsifiers of ϕ .*

The subject matter of ϕ $sm(\phi)$ is a state in S such that $sm(\phi) = \sqcup \mathcal{V}(\phi) \sqcup \sqcup \mathcal{F}(\phi)$.¹

2 Simple Frameworks

With the materials from truthmaker semantics, now I develop two simple frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$ that immediately arise from the idea that the expansion and contraction operates on a belief-

¹The standard definition of subject matter is in terms of *propositions* defined in terms of states: a *bilateral proposition* P is defined as $\langle P^+, P^- \rangle$ where P^+ and P^- contains its verifiers and falsifiers respectively. Then the subject matter of P is either the pair $\langle \sqcup P^+, \sqcup P^- \rangle$ (*differentiated subject matter*) or the fusion of $\sqcup P^+$ and $\sqcup P^-$ (*comprehensive subject matter*). I intentionally deviated from the standard definition for the project of this paper the commitment for the theory of truthmaker content is not required. The preference for the comprehensive over the differentiated subject matter in epistemic settings follows Jago (2024).

state set $\mathcal{S}^{\mathcal{K}}$. Two simple frameworks will only differ in how to determine the belief set $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$. Thus the definitions in this section apply to both frameworks if not otherwise noted.

In simple frameworks we do not pose any constraint on $\mathcal{S}^{\mathcal{K}}$:

Definition 2.1 (Simple Belief-State Set). *Given a model $M = \langle S, \mathcal{S}^\diamond, \sqsubseteq, |\cdot| \rangle$, a belief-state set $\mathcal{S}^{\mathcal{K}}$ is a subset of S .*

In simple frameworks, an expansion of $\mathcal{S}^{\mathcal{K}}$ with a state σ ($\mathcal{S}^{\mathcal{K}} + \sigma$) simply adds σ to $\mathcal{S}^{\mathcal{K}}$ if not already contained; and a contraction of $\mathcal{S}^{\mathcal{K}}$ with σ ($\mathcal{S}^{\mathcal{K}} \div \sigma$) simply extracts σ from $\mathcal{S}^{\mathcal{K}}$ only if it was originally contained. Thus:

Definition 2.2 (Expansion and Contraction in Simple Frameworks). *Given a belief-state set $\mathcal{S}^{\mathcal{K}}$, $\mathcal{S}^{\mathcal{K}} + \sigma = \mathcal{S}^{\mathcal{K}} \cup \{\sigma\}$; and $\mathcal{S}^{\mathcal{K}} \div \sigma = \mathcal{S}^{\mathcal{K}} \setminus \{\sigma\}$.*

As the ontology of states does not matter for our purpose, the ontology of belief-state set is also not required. Yet intuitively, one can regard $\mathcal{S}^{\mathcal{K}}$ as a set of states that the modelled epistemic agent is epistemically committed into; that is, the set of sources of all its beliefs. My preferred understanding is to take $\mathcal{S}^{\mathcal{K}}$ as a model for the modelled epistemic agent's 'representation' of reality, in the broadest sense. Thus, the belief set would contain every formula that would be actually verified if the representation were totally correct. Since we have two different notions of verification: exact and inexact, we would have two different frameworks that decide $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$:

Definition 2.3 (Belief Set in $\mathcal{E}_0$ and $\mathcal{I}_0$). *In framework $\mathcal{E}_0$, given a simple belief-state set $\mathcal{S}^{\mathcal{K}}$, a belief set $\mathcal{K} = \{\phi : \exists \sigma \in \mathcal{S}^{\mathcal{K}} \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.*

In framework $\mathcal{I}_0$, given a simple belief-state set $\mathcal{S}^{\mathcal{K}}$, a belief set $\mathcal{K} = \{\phi : \exists \sigma \in \mathcal{S}^{\mathcal{K}} \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.

Notice that two basic frameworks only differ in this respect. We could define a framework as an n -tuple. Thus:

Definition 2.4 (Frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$). *Given a model M , the framework $\mathcal{E}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{E}} \rangle$ where Σ_0 is a set of belief-state set $\{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S\}$, $+$ and $\div$ are expansion*

and contraction respectively, and Δ_s are decision function from the set of belief-state sets to the set of belief sets such that $\Delta_{\mathcal{E}}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.

Given a model M , the framework $\mathcal{I}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{I}} \rangle$ where $\Delta_{\mathcal{I}}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$, and everything else is as defined above.

One might desire there to be additional notions of expansion and contraction that operates on $\mathcal{K}$. Intuitively, the expansion of $\mathcal{K}$ with a formula ϕ ($\mathcal{K} \oplus \phi$) would be the resulting belief set of the belief-state set $\mathcal{S}^{\mathcal{K}}$ expanded with *some* verifiers of ϕ ; the contraction of $\mathcal{K}$ with a formula ϕ ($\mathcal{K} \ominus \phi$) would be the resulting belief set of the belief-state set $\mathcal{S}^{\mathcal{K}}$ contracted with *all* verifiers of ϕ .

However, it should be noticed that given a definite $\mathcal{S}^{\mathcal{K}}$, the resulting belief set $\mathcal{K}$ is definite in both $\mathcal{E}_0$ and $\mathcal{I}_0$; whereas given a definite belief set one cannot track down to a unique belief-state set. So let us denote the particular belief set decided by a belief-state set $\mathcal{S}^{\mathcal{K}}$ $D(\mathcal{S}^{\mathcal{K}})$. More specifically, the belief state decided by a belief-state set $\mathcal{S}^{\mathcal{K}}$ in a framework X will be denoted by $D_X(\mathcal{S}^{\mathcal{K}})$. The specification of the framework will be omitted unless context is insufficient to disambiguate. Then, the expansion and contraction for belief set should be defined as follows:

Definition 2.5 (Operations on Belief Set in Simple Frameworks). *Given a belief-state set $\mathcal{S}^{\mathcal{K}}$ and a selection function γ ,*

$$(\mathcal{E}_0^{\oplus}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \oplus \phi = \gamma(D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}} + \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{E}_0^{\ominus}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{E}_0} \phi = D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}} \setminus \{\sigma : \sigma \Vdash \phi\})$$

$$(\mathcal{I}_0^{\oplus}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \oplus \phi = \gamma(D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}} + \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{I}_0^{\ominus}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{I}_0} \phi = D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}} \setminus \{\sigma : \sigma \Vdash \phi\})$$

The revision on belief set ($\circledast$) is defined via Levi Identity:

$$(\mathcal{E}_0^{\circledast}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \circledast_{\mathcal{E}_0} \phi = (D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{E}_0} \neg\phi) \oplus \phi$$

$$(\mathcal{I}_0^{\circledast}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \circledast_{\mathcal{I}_0} \phi = (D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{I}_0} \neg\phi) \oplus \phi$$

This yields the result that still expansion, and thus revision on belief set is not definite in $\mathcal{E}_0$

nor in $\mathcal{I}_0$, for a formula can have multiple verifiers, exact or inexact. But I got an excuse for that. As I said, we can think $\mathcal{S}^\mathcal{K}$ to be the set of all doxastic sources, or reason the modelled epistemic agent have upon forming beliefs. So it seems natural to me that, given a belief-state set, one cannot determine what would be the result after one acquired a belief, for it is not specified what is the exact source of that acquired belief. However, the definition also allows the expansion infringe *minimal action* for expansion (see Table 1). This feature, I admit, may be disputed. One might add the clause “if $\phi \in D(\mathcal{S}^\mathcal{K})$, $D(\mathcal{S}^\mathcal{K}) \oplus \phi = D(\mathcal{S}^\mathcal{K})$ ” to the definition to retain the minimal action postulate, if one wishes. On the contrary, it seems given a belief-state set, we could determine what would be the result after one discarded a belief: the agent would have lost *all* the sources for the discarded belief. Also notice that in case of contraction, *minimal action* for contraction (see Table 2) holds.

Now let us observe how two simple frameworks satisfies every desideratum except **Local Rationality**.

(D1) **Definite Contraction**

It would be easy to check that the contraction (on belief-state set) is uniquely determined, since relative complement is uniquely determined.

One interesting feature of these frameworks is that we can now define *entrenchment* of the belief that $\phi \in D(\mathcal{S}^\mathcal{K})$ by inspecting $\mathcal{S}^\mathcal{K}$. As for verification, we have exact and inexact entrenchment, such that:

Definition 2.6 ((Entrenchment)). *Given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$,*

The exact entrenchment of ϕ is the number of exact verifiers of ϕ in $\mathcal{S}^\mathcal{K}$, i.e., $\text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$

The inexact entrenchment of ϕ is the number of inexact verifiers of ϕ in $\mathcal{S}^\mathcal{K}$, i.e., $\text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$

The intuitive idea behind is that the modelled epistemic agent is more entrenched to the belief that ϕ just in case it has more sources or reason of believing that ϕ . Observe that different frameworks get along well with different notions of entrenchment: the natural notion of en-

trenchment in framework $\mathcal{E}_0$ is *exact entrenchment*, while entrenchment in framework $\mathcal{I}_0$ it is *inexact entrenchment*.

With the notion of entrenchment, one thing we can now make sense is the phenomenon that sometimes expanding or contracting on $\mathcal{S}^\mathcal{K}$ which makes difference on the belief-state set may still not result in any difference on $\mathcal{K}$. Suppose the state $t \notin \mathcal{S}^\mathcal{K}$ only verifies ϕ , but $s \in \mathcal{S}^\mathcal{K}$ such that $s \models \phi$, and all fusions of t and any state in $\mathcal{S}^\mathcal{K}$ is not in $\mathcal{S}^\mathcal{K}$. Then $\mathcal{S}^\mathcal{K} + t$ would make no difference in $\mathcal{K}$, under both frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$; likewise $\mathcal{S}^\mathcal{K} \cup \{t\} \div t$ would make no difference in $\mathcal{K}$. This, I submit, can be made sense by the notion of entrenchment we just defined. Although the belief set remains the same, by expanding with t the entrenchment of ϕ has increased, i.e., the belief that ϕ is reinforced. Likewise, by contracting with t the entrenchment of ϕ has decreased, i.e., the belief that ϕ is weakened.

(D2) **Logical Ignorance**

This is relatively easy to show. In fact, it is rather difficult to construct a belief-state set that makes all logical truths contained in the belief set. To establish logical omniscience, the belief-state set must include, for any formula, either the verifier of it or the falsifier of it. Otherwise there are some logical truths that is not a member of $\mathcal{K}$. One extreme example is where $\mathcal{S}^\mathcal{K} = \emptyset$, which is allowed under the definition. Then by definition for belief set, $\mathcal{K}$ would contain no formula at all, needless to say all logically true formulas.

(D3) **Paraconsistency**

The fact that the frameworks satisfy **Paraconsistency** is also very much direct. Notice that in simple frameworks there are no constraint on $\mathcal{S}^\mathcal{K}$ that it should not contain both verifiers and falsifiers for some formulas. Thus, suppose $\mathcal{S}^\mathcal{K} = \{s, t\}$ where s verifies (exactly for $\mathcal{E}_0$, inexactly for $\mathcal{I}_0$. Similarly, for below) ϕ and t falsifies ϕ (therefore verifies $\neg\phi$); and each state in $\mathcal{S}^\mathcal{K}$ verifies or falsifies nothing else. Then $D(\mathcal{S}^\mathcal{K}) = \{\phi, \neg\phi\}$, without introducing all formulas in $\mathcal{L}$ into $D(\mathcal{S}^\mathcal{K})$.

However, this maybe thought also as a weakness of simple frameworks, for it means that the frameworks allow blatant contradictions. The details will be discussed below when discussing

Local Rationality.

(D5) Hyperintensionality

Hyperintensionality is automatically granted in our frameworks, for underlying truthmaker model is already hyperintensional. Suppose $s_1 \Vdash p$ and $s_2 \nVdash p$; $t_1 \Vdash q$ and $t_2 \nVdash q$. What follows is that although $\vdash p \vee \neg p$ and $\vdash q \vee \neg q$, they may have different verifiers: for $s_1 \Vdash p \vee \neg p$ and $t_1 \Vdash q \vee \neg q$. Likewise, although $p \wedge \neg p$ and $q \wedge \neg q$ are all logical falsities, they may have different verifiers: for $s_1 \sqcup s_2 \Vdash p \wedge \neg p$ and $t_1 \sqcup t_2 \Vdash q \wedge \neg q$. Recall that there were no constraints on model $\mathcal{M}$ that any states must verify logical truths or falsify logical falsities. Thus expanding or contracting with $p \vee \neg p$ on $\mathcal{K}$ is not equivalent with expanding or contracting with $q \vee \neg q$; likewise expanding or contracting with $p \wedge \neg p$ on $\mathcal{K}$ is not equivalent with expanding or contracting with $q \wedge \neg q$.

(D6) Pertinent Recovery

Recall that the problematic case of recovery: ϕ strictly implies ψ , so contracting with ψ would result in removing ϕ from $\mathcal{K}$; yet by *recovery* (in Table 2) expanding with ψ would recover not only ψ but also ϕ . However, this contradicts with our daily observations, for the modelled epistemic agent could believe ψ with some sources that do not force it to believe ϕ .

The proposed frameworks deal with this kind of problem particularly well. Let us suppose all verifiers of ϕ , including s , are also verifiers of ψ , but there is a state t which verifies ψ but not ϕ . By supposition, if $s \in \mathcal{S}^{\mathcal{K}}$, $\mathcal{K}$ will include both ϕ and ψ ; and $\{\phi, \psi\} \cap (D(\mathcal{S}^{\mathcal{K}}) \ominus \psi) = \emptyset$. But still, it is not necessarily that $\phi \in (D(\mathcal{S}^{\mathcal{K}}) \ominus \psi) \oplus \psi$. Since expansion on a belief set is not uniquely determined, $(D(\mathcal{S}^{\mathcal{K}}) \ominus \psi) \oplus \psi$ might be the set that contains the state s that verifies both ϕ and ψ , but might as well be the set that contains the state t that does not verify ϕ , and thus $\phi \notin (D(\mathcal{S}^{\mathcal{K}}) \ominus \psi) \oplus \psi$.

So the idea that, given merely the assumptions that ϕ entails ψ and $\{\psi, \phi\} \subseteq \mathcal{K}$ we cannot know whether $(\mathcal{K} \ominus \psi) \oplus \psi$ includes ϕ is preserved. We of course, can know whether s is in $(\mathcal{S}^{\mathcal{K}} \div t) + t$ given the information whether $\{s\} \subseteq \mathcal{S}^{\mathcal{K}}$. But as noted in §1.1.4, the recovery formulated in terms of the source of belief - states and belief-state set - is extremely compelling.

To sum up, two simple frameworks - $\mathcal{E}_0$ and $\mathcal{I}_0$ already do very well in satisfying all the desiderata except **Local Rationality**. Yet they are untenable for they fail to satisfy the only remaining desideratum so drastically.

2.1 Rationality

As mentioned above, because there were no constraints on $\mathcal{S}^{\mathcal{K}}$ in both simple frameworks, they allow $\{\phi, \neg\phi\} \subseteq \mathcal{K}$, and also $\phi \wedge \neg\phi \in \mathcal{K}$ without any tension. Although I do not submit that the modelled agent should never be capable of having an inconsistent belief set, I do hold that inconsistent belief set must cause some sort of ‘tension’. This is required if one wants to have a suitable notion of revision on $\mathcal{S}^{\mathcal{K}}$.

Recall that in simple frameworks revision is defined only on a belief set. This is because without any constraints on consistency of $\mathcal{S}^{\mathcal{K}}$, any attempts to define revision would be arbitrary, or too strong to satisfy other desiderata. For instance, one might wish to define revising with σ on $\mathcal{S}^{\mathcal{K}}$ ($\mathcal{S}^{\mathcal{K}} * \sigma$) to be $(\mathcal{S}^{\mathcal{K}} \setminus \{\tau : \tau \sqcup \sigma \notin S^{\diamond}\}) \cup \{\sigma\}$. The idea is that, if we were to revise with σ , we first contract with all states that is *incompatible* with σ , and then expand with σ . Then by definition of a model, $D(\mathcal{S}^{\mathcal{K}} * \sigma)$ will exclude all the formulas that is exactly falsified by σ . So far so good, but this also excludes all formulas in $D(\mathcal{S}^{\mathcal{K}})$ such that every verifier of it is an impossible state; i.e., a logical falsity, for $\tau \sqcup \sigma \notin S^{\diamond}$ if $\tau \notin S^{\diamond}$. So after a revision on belief-state set with σ it would result in removing all logical falsities. This would infringe our desideratum on **Hyperintensionality**. What if σ already is an impossible state? Actually the result is much more devastating: if $\sigma \notin S^{\diamond}$, according to the proposed definition on revision, $\mathcal{S}^{\mathcal{K}} * \sigma = \{\sigma\}$, for any τ will fuse with σ to an impossible state. Since we wanted the modelled agent to have minimal changes even it is forced to believe inconsistently, we might want to say this violates **Paraconsistency**.

Another unfortunate feature of simple frameworks is that *every* belief must be incorporated to the agent manually. $\mathcal{I}_0$ is perhaps better in this respect for by believing a conjunction it forces the agent to believe each of its conjuncts, while $\mathcal{E}_0$ does not yield this feature since the exact verifier of a conjunction is not generally the exact verifier of its conjuncts. But it should

be examined whether this feature is really desirable. I postpone this discussion to §2.3.

One especially problematic consequence of both basic frameworks is that one can believe that ϕ and $\phi \rightarrow \psi$ ($\iff \neg\phi \vee \psi$, since $\rightarrow$ is not a primitive operator in our language), yet not believing ψ . For the reason I will suggest in the next subsection, I admit there can be cases where one believes both $\phi \wedge \chi$ and $\phi \rightarrow \psi$, but not ψ . But it seems to me, a locally rational epistemic agent must believe that ψ , given that the agent believes that ϕ , and it also believes that either it is the case that $\neg\phi$ or ψ . In short, we want the modelled epistemic agent rational enough to have a belief set closed under modus ponens.

To sum up, simple frameworks fail to satisfy the ‘rationality’ part of **Local Rationality**, for it permits blatant inconsistency in $\mathcal{K}$, and also because it fails to make the belief set closed under modus ponens (of material conditional). One might additionally complain that it is not closed under conjunction elimination. It is, for some parts, too weak to satisfy the desideratum. However, it is sometimes too strong to satisfy the ‘locality’ part of **Local Rationality**.

2.2 Locality

In some sense, it may be said that to model the locality of the rationality of the modelled epistemic agent is more important to our present purpose. For, if one only wanted the belief set to be gently consistent, or closed under modus ponens or conjunction elimination, the traditional AGM is already sufficient. In the previous section, we observed some cases that the frameworks fail to yield even local rationality. But there are also cases where the frameworks yield consequences than what we might desire.

One crucial failure of locality is that in both simple frameworks, the belief set is closed under disjunction introduction: if $\phi \in \mathcal{K}$, for any formula ψ , $\phi \vee \psi \in \mathcal{K}$. It is easy to notice that we do not believe this way: although Frege would have believed that Wilhelm II is the King of Prussia, he would have not believed that Wilhelm II is the King of Prussia or Friedrich Merz is the chancellor of Germany.

Besides intuitive implausibility, the closure under disjunction introduction also has further problems. For one thing, this makes any modelled epistemic agent believe *about* everything.

Let us define the subject matter of a *set* of formulas X as the least upper bound of the set of all subject matter of the formulas in X , formally:

Definition 2.7 (Subject Matter for Sets). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$, the subject matter of a set $X \subseteq \mathcal{L}$ is defined as $sm(X) = \sqcup \{ \sigma : \exists \phi \in X \text{ s.t. } sm(\phi) = \sigma \}$.*

With the notion of subject matter for sets, we might want to investigate what the modelled epistemic agent believes *about*. One intriguing example is that we can now define a stronger notion of ignorance: the agent is ignorant about ϕ if $\{ \phi, \neg \phi \} \cap \mathcal{K} = \emptyset$; but the agent is strongly ignorant about ϕ if the only common part of $sm(\phi)$ and $sm(\mathcal{K})$ is $\square$.

Yet if $\mathcal{K}$ is closed under disjunction introduction, there are no formulas that the agent is strongly ignorant. In fact, for any possible $\mathcal{K}$ that is not empty, $sm(\mathcal{K}) = sm(\mathcal{L})$ under both simple frameworks. This is clearly not desirable. It seems that until I searched on the internet to write this paragraph, I was strongly ignorant about that Amr ibn al-Layth was an Amir of Saffarid dynasty. But simple frameworks deny this intuition, for since I believe that I have a hand, my belief-state set would include a state s that verifies the statement “I have a hand”, which also verifies “I have a hand or Amr ibn al-Layth was an Amir of Saffarid dynasty”. And since the subject matter of the statement “Amr ibn al-Layth was an Amir of Saffarid dynasty” is a part of the subject matter of the disjunction, I would be not strongly ignorant about the statement about Amr ibn al-Layth, which I never have heard before.

Thus, it would be better to have a belief set that is not closed under disjunction introduction. However, excluding the disjunctive consequence of beliefs should be carefully executed; for we do not wish to ban the modelled agent to have disjunction as its belief. Notice that any verifiers, exact or inexact, of a disjunction $\phi \vee \psi$ must be a verifier, exact or inexact, of at least one of its disjuncts ϕ or ψ . So if we restrict the modelled epistemic agent believe $\phi \vee \psi$ on the basis of the belief that ϕ , the result is that none of the member of $\mathcal{K}$ is a disjunction. This suggests that we may need a more fine-grained constraint on locality.

2.3 Closure Under Conjunction Elimination?

Finally, let me also note a problem about conjunction. Notice that in $\mathcal{E}_0$ even if $\phi \wedge \psi \in \mathcal{K}$, it does not grant that $\{\phi, \psi\} \subseteq \mathcal{K}$, whereas in $\mathcal{I}_0$ it is guaranteed. So at the first sight, it may seem as a reason to favor $\mathcal{I}_0$ over $\mathcal{E}_0$. One might therefore conclude that $\mathcal{K}$ should be defined from $\mathcal{S}^{\mathcal{K}}$ via the notion of *inexact* verification rather than the notion of *exact* verification.

The idea that the belief set should be closed under conjunction elimination seems plausible. However, I do not wish to adjudicate which notion of verification is suitable for defining $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$. The worry of using the exact notion of verification is as stated: the belief set can contain a conjunction without containing any of its conjuncts, which might make the modelled agent seem less rational than we wish it to be. Indeed, there might be something wrong with an agent if it believes $\phi \wedge \psi$ without believing that ϕ or ψ .

But using the inexact notion of verification is not without problems. We say ϕ is a conjunctive part of ψ (and ψ is a conjunctive whole of ϕ) iff: (a) for all $s \Vdash \psi$, there is a t such that $t \sqsubseteq s$ and $t \Vdash \phi$, and (b) for all $t \Vdash \phi$, there is an s such that $t \sqsubseteq s$ and $s \Vdash \psi$ (and (c) any falsifier of ϕ is a falsifier of ψ . Clause (c) may be omitted sometimes)². Notice that the definition for conjunctive part does not involve the syntactic structure of ϕ or ψ . Now, under $\mathcal{I}_0$ not only each conjunct of a conjunction in $\mathcal{K}$ is included in $\mathcal{K}$, but all *conjunctive part* of a member of $\mathcal{K}$ is included in $\mathcal{K}$. This is direct from the fact that any inexact verifier of a conjunctive whole would be an inexact verifier of its conjunctive part.

But it is unclear why should the modelled epistemic agent must believe all its conjunctive parts of its belief, especially when the conjunctive part in question is not syntactically related to its whole. At least in some cases, it seems there are good reason to believe that believing the conjunctive whole give sufficient condition to believe its conjunctive part regardless of their syntax. For example, it is reasonable to think “ x is red” is a conjunctive part of “ x is scarlet”; and if one believes that x is scarlet, one has sufficient reason to believe that x is red (Jago, 2024).

But it should also be noticed that sometimes the notion of conjunctive parthood is used to

²Again, the notion of conjunctive part is also usually defined in terms of propositions; but let us operate with a watered-down notion of conjunctive part of formulas.

formulate semantic or metaphysical notions. It is customary to think if ϕ is a conjunctive part of ψ , the proposition that ϕ is a *partial content* of the proposition that ψ (Fine 2016, 2017a, 2017b). And it is also suggested that ϕ is a strict partial ground of ψ iff ϕ is a conjunctive part of ψ but not vice versa (Fine, 2012); or that ϕ a partial essence of ψ iff ϕ is a conjunctive part of ψ (Correia & Skiles, 2017). If these suggestions are correct, defining $\mathcal{K}$ to contain every formula inexactly verified by some members of $\mathcal{S}^{\mathcal{K}}$ would result in the modelled agent to believe every partial contents, every strict partial grounds, and every partial essence of its beliefs - which sounds very suspicious. Of course, none of these suggestions are unchallenged; but $\mathcal{I}_0$ or its kin would be just as suspicious as those accounts are compelling.

Still, I personally favor to define $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$ by inexact verification. At least the semantic application of conjunctive parthood does not seem to threaten $\mathcal{I}_0$, for it may be claimed reasonable to expect an ideal epistemic agent should not believe ϕ if it is not fully aware of its semantic content. Nevertheless, the consideration above shows that frameworks that define $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$ by exact verification also deserve substantial interest. Also, as we will see in the next section, the approach also validates more fine-grained distinction on the constraints on $\mathcal{K}$. For such reasons both approaches will be developed in the remainder of the paper.

A final remark. There is in fact a moderate alternative between exact and inexact approaches: given a model M and a belief-state set $\mathcal{S}^{\mathcal{K}}$, we first define $\mathcal{K}_0$ to be the set of all formulas *exactly* verified by some members of $\mathcal{S}^{\mathcal{K}}$; $\mathcal{K}_{n+1}$ is defined as $\mathcal{K}_n \cup \{\phi, \psi : \phi \wedge \psi \in \mathcal{K}_n\}$; and finally the belief set $\mathcal{K}$ is defined as $\bigcup_{n=0}^{\infty} \mathcal{K}_n$. This yields the result that $\mathcal{K}$ is closed under conjunction elimination, but not necessarily closed under conjunctive parthood. I will not develop this kind of approach further in this paper, but this is indeed an option.

3 More Frameworks!

3.1 The Locality Constraint

Let us solve the problems of simple frameworks step-by-step. It turns out that the problem of disjunction is much easier to be dealt with. We only modify the definition of belief set to get

the simple frameworks supplemented with the locality constraint:

Definition 3.1 (Frameworks $\mathcal{EL}_0$). *Given a model M , the framework $\mathcal{EL}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{EL}} \rangle$ where $\Delta_{\mathcal{EL}}(X) = \{ \phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi \ \& \ \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \Box(\tau \sqsubseteq \text{sm}(\psi) \ \& \ \tau \sqsubseteq \sqcup X) \} \subseteq \mathcal{L}$, and everything else is as defined in **Definition 2.4**.*

Definition 3.2 (Frameworks $\mathcal{IL}_0$). *Given a model M , the framework $\mathcal{IL}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{IL}} \rangle$ where $\Delta_{\mathcal{IL}}(X) = \{ \phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi \ \& \ \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \Box(\tau \sqsubseteq \text{sm}(\psi) \ \& \ \tau \sqsubseteq \sqcup X) \} \subseteq \mathcal{L}$, and everything else is as defined in **Definition 2.4**.*

The expansion and contraction with formulas under the framework $\mathcal{EL}_0$ is defined as it was defined under $\mathcal{E}_0$; and similarly under $\mathcal{IL}_0$ with $\mathcal{I}_0$.

The only differences of these two new frameworks are that we have an additional locality constraints on deciding $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$: intuitively, the subject matter of formulas in $\mathcal{K}$ must have something common of $\mathcal{S}^{\mathcal{K}}$. Given $\phi \in \mathcal{K}$, the locality constraint yields that $\phi \vee \psi \in \mathcal{K}$ if and only if $\text{sm}(\psi)$ has a non-trivial common part with $\sqcup \mathcal{S}^{\mathcal{K}}$. Roughly, by incorporating the locality constraint on Δ , we allow the modelled epistemic agent to have disjunctive belief, but only to the extent that each of the disjuncts are “partially entertained in the representation”, positively or negatively.

This restores the intuition that epistemic agents can be strongly ignorant of some formulas. Suppose $\mathcal{S}^{\mathcal{K}} = \{s_1, s_2\}$, $|p|^+ = \{t_1\}$ and $|p|^- = \{t_2\}$, and $s_1 \Vdash q$ and $s_2 \dashv\vdash r$. Finally, suppose $\text{sm}(p) (= t_1 \sqcup t_2) \not\sqsubseteq \sqcup \mathcal{S}^{\mathcal{K}} (= s_1 \sqcup s_2)$. Then, under both frameworks $\mathcal{EL}_0$ and $\mathcal{IL}_0$, $\{q \vee r, q \vee \neg r\} \subseteq \mathcal{K}$, but $\{q \vee p, q \vee \neg p\} \cap \mathcal{K} = \emptyset$. Furthermore, if q nor r has its exact verifiers or falsifiers which has t_1 or t_2 as part, and there are no other atomic formulas, we could conclude that the agent which has such a belief set is strongly ignorant about p , i.e., the only common part of $\text{sm}(p)$ and $\text{sm}(\mathcal{K})$ is $\Box$.

3.2 Consistency Constraints

By contrast, the problem of rationality is difficult - not because it requires a highly complex formal mechanism to solve, but because there are so many different degrees of rationality one

might want to impose on the modelled epistemic agent.

One main problem of simple frameworks addressed in §2.1 was that simple frameworks allow the belief set to consist both ϕ and $\neg\phi$. However, if a given $\mathcal{S}^\mathcal{K}$ already consists of both verifier and falsifier of a formula ϕ (thus verifiers for ϕ and $\neg\phi$), it would be arbitrary to choose one from ϕ or $\neg\phi$ to include in $\mathcal{K}$. To use the metaphor of “source”: if an epistemic agent has both sources, or reasons of believing that ϕ and believing that $\neg\phi$, it would seem incorrect to ascribe only one of the belief to the agent. Instead, since we want to model a somewhat ideal epistemic agent, not all sorts of bizarre believer, we could simply forbid the modelled epistemic agent to have both verifier and falsifier for any formula ϕ in its “source”.

This idea immediately suggests some constraints we can impose on the definition of $\mathcal{S}^\mathcal{K}$. Since we have two notions of verification and falsification, there are in fact more than one constraint arise from the idea above.

Definition 3.3 (Set of Belief-State Sets with Consistency Constraints). *Given a model M ,*

$$(0) \ \Sigma_0 = \{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S\}$$

$$(1) \ \Sigma_1 = \{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^\mathcal{K}, \neg\exists\phi (s \Vdash \phi \ \& \ t \Vdash \neg\phi)\}$$

$$(2a) \ \Sigma_{2a} = \{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^\mathcal{K}, \neg\exists\phi (s \Vdash \phi \ \& \ t \Vdash \phi)\}$$

$$(2b) \ \Sigma_{2b} = \{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^\mathcal{K}, \neg\exists\phi (s \Vdash \neg\phi \ \& \ t \Vdash \neg\phi)\}$$

$$(3) \ \Sigma_3 = \Sigma_{2a} \cap \Sigma_{2b}$$

$$(4) \ \Sigma_4 = \{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^\mathcal{K}, \neg\exists\phi (s \Vdash \phi \ \& \ t \Vdash \neg\phi)\}$$

It would be easy to check that $\Sigma_4 \subseteq \Sigma_3 \subseteq \Sigma_{2a} \subseteq \Sigma_1 \subseteq \Sigma_0$, and $\Sigma_4 \subseteq \Sigma_3 \subseteq \Sigma_{2b} \subseteq \Sigma_1 \subseteq \Sigma_0$.

Before we proceed to define our new frameworks utilizing consistency constraints, we need to first refine our definition of $+$. If we continue to regard $\mathcal{S}^\mathcal{K} + \sigma = \mathcal{S}^\mathcal{K} \cup \{\sigma\}$, the result of expansion would not generally be a belief-state set, i.e., member of a Σ_n , since $\mathcal{S}^\mathcal{K}$ may contain a falsifier of a formula that σ verifies (or vice versa). The modification, however, is rather simple, for we only need to add a condition for an expansion operation to succeed, as there was a condition for a contraction operation to success in the original AGM framework.

Definition 3.4 (Expansion and Contraction in General). *Given a belief-state set $\mathcal{S}^\mathcal{K} \in \Sigma_n \subseteq \Sigma_0$ and $\sigma \in S$,*

$$\mathcal{S}^\mathcal{K} +_n \sigma = \begin{cases} \mathcal{S}^\mathcal{K} \cup \{\sigma\} & \text{if } \mathcal{S}^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ \mathcal{S}^\mathcal{K} & \text{otherwise} \end{cases}$$

$$\mathcal{S}^\mathcal{K} \div \sigma = \mathcal{S}^\mathcal{K} \setminus \{\sigma\}$$

With the new conception of expansion with states, we can now further define a suitable notion of revision with states. The idea is that, if one revise with σ on $\mathcal{S}^\mathcal{K}$, we first contract minimal amount of state from $\mathcal{S}^\mathcal{K}$ to make the expansion with σ successful, and then expand with σ with the result of the contractions. First, we define conflict relations Θ that correspond with each consistency constraints:

Definition 3.5 (Conflict Relations Corresponding to Consistency Constraints). *Given a model M and σ, τ both in S ,*

$$(0) \quad \forall \sigma, \tau \neg \Theta_0(\sigma, \tau)$$

$$(1) \quad \Theta_1(\sigma, \tau) \iff \exists \phi ((\sigma \Vdash \phi \ \& \ \tau \dashv \Vdash \phi) \vee (\tau \Vdash \phi \ \& \ \sigma \dashv \Vdash \phi))$$

$$(2a) \quad \Theta_{2a}(\sigma, \tau) \iff \exists \phi ((\sigma \Vdash \phi \ \& \ \tau \lhd \Vdash \phi) \vee (\tau \Vdash \phi \ \& \ \sigma \lhd \Vdash \phi))$$

$$(2b) \quad \Theta_{2b}(\sigma, \tau) \iff \exists \phi ((\sigma \rhd \Vdash \phi \ \& \ \tau \dashv \Vdash \phi) \vee (\tau \rhd \Vdash \phi \ \& \ \sigma \dashv \Vdash \phi))$$

$$(3) \quad \Theta_3(\sigma, \tau) \iff \Theta_{2a}(\sigma, \tau) \vee \Theta_{2b}(\sigma, \tau)$$

$$(4) \quad \Theta_4(\sigma, \tau) \iff \exists \phi ((\sigma \rhd \Vdash \phi \ \& \ \tau \lhd \Vdash \phi) \vee (\tau \rhd \Vdash \phi \ \& \ \sigma \lhd \Vdash \phi))$$

Then we can define revision with states via Quasi-Levi-Identity:

Definition 3.6 (Revision with Consistency Constraint). *Given a model M and $\sigma \in S$*

$$\mathcal{S}^\mathcal{K} *_n \sigma = (\mathcal{S}^\mathcal{K} \setminus \{\tau : \Theta_n(\sigma, \tau)\}) \cup \{\sigma\}.$$

That is, let $\langle \tau_i \rangle_{i=1}^k$ be an arbitrary enumeration of all states in $\mathcal{S}^{\mathcal{K}}$ such that $\Theta_n(\sigma, \tau_i)$ holds. We construct a sequence of contractions where $\mathcal{S}_0 = \mathcal{S}^{\mathcal{K}}$ and $\mathcal{S}_{i+1} = \mathcal{S}_i \div \tau_i$. The revision operation is then defined purely in terms of contraction and expansion as:

$$\mathcal{S}^{\mathcal{K}} *_n \sigma = \mathcal{S}_k +_n \sigma$$

Now we are ready to define new frameworks using consistency constraint. In principle, we could compose numerous frameworks with the resources given, but only four of them would worth closer examination:

Definition 3.7 (Frameworks $\mathcal{EL}_1$, $\mathcal{EL}_3$, $\mathcal{EL}_4$, and $\mathcal{IL}_4$). *Given a model M ,*

The framework $\mathcal{EL}_1$ is a quintuple $\langle M, \Sigma_1, +_1, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{EL}_3$ is a quintuple $\langle M, \Sigma_3, +_3, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{EL}_4$ is a quintuple $\langle M, \Sigma_4, +_4, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{IL}_4$ is a quintuple $\langle M, \Sigma_4, +_4, \div, \Delta_{\mathcal{IL}} \rangle$.

These (perhaps except $\mathcal{EL}_1$) are the first frameworks presented in the paper which satisfy all the desiderata in §1.1. Notice that any $\mathcal{IL}$ -series frameworks which have weaker consistency constraint than that of Σ_4 would still suffer from blatant inconsistency ($\{\phi, \neg\phi\} \subseteq \mathcal{K}$). Suppose $s \Vdash p$ and $t \Vdash \neg p$, but p is not exactly verified or falsified by any of them. Furthermore, suppose s and t does not exactly verify or falsify any formula. Then under consistency constraint weaker than that of Σ_4 it could permit $\{s, t\} \subseteq \mathcal{S}^{\mathcal{K}}$, and by the definition of $\Delta_{\mathcal{IL}}$, it could therefore permit $\{p, \neg p\} \subseteq \mathcal{K}$. However, obviously, $\mathcal{EL}_1$ would not suffer from the same problem.

Interestingly, the consistency constraint can also guarantee the belief set to be closed under modus ponens. For convenience, I will only show $\mathcal{EL}_1$ satisfy this desideratum, and it will be obvious that other three frameworks also have this feature. We assume the contrary, that $\{\phi, \neg\phi \vee \psi\} \subseteq \mathcal{K}$ but $\psi \notin \mathcal{K}$. This implies that $\mathcal{S}^{\mathcal{K}}$ contains some exact verifier of ϕ and do not contain any exact verifier of ψ ; and by consistency constraint it follows that $\mathcal{S}^{\mathcal{K}}$ do not contain any exact falsifier of ϕ . However, since any exact verifier of $\neg\phi \vee \psi$ is either an exact falsifier

of ϕ or an exact verifier of ψ . And for nothing above is in $\mathcal{S}^{\mathcal{K}}$, $\neg\phi \vee \psi \notin \mathcal{K}$, contradiction. Therefore, for any formulas ϕ and ψ , if $\{\phi, \neg\phi \vee \psi\} \subseteq \mathcal{K}$ then $\psi \in \mathcal{K}$, i.e., $\mathcal{K}$ is closed under modus ponens.

The difference between three new $\mathcal{EL}$ -series frameworks can be appreciated by examining which set of formulas are possible subset of $\mathcal{K}$ under different frameworks. For example, we can observe that if one say $\mathcal{EL}_1$ satisfy **Local Rationality**, one is saying in a quite weak sense of rationality, since it is possible that $\{p \wedge \neg p\} \subseteq \mathcal{K}$, or $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$. Suppose s_1 and s_2 are sole exact verifier and falsifier of p respectively, and t_1 and t_2 are sole exact verifier and falsifier of q respectively. Let us further suppose no mentioned state is a part of another mentioned state, and p and q are only atomic formulas. Then necessary and sufficient condition for $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$ to be the case under $\mathcal{EL}_1$ is that $\{s_1 \sqcup t_1, s_2 \sqcup t_2\} \subseteq \mathcal{S}^{\mathcal{K}}$. It should be noticed that there are no formulas *exactly* verified by $s_1 \sqcup t_1$ yet *exactly* falsified by $s_2 \sqcup t_2$, and also vice versa: atomic formulas p or q (and therefore $\neg p$ or $\neg q$) are merely inexactly verified/falsified by them; $p \vee q$ (and therefore $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$) is exactly falsified (verified) by $s_2 \sqcup t_2$ yet merely inexactly verified (falsified) by $s_1 \sqcup t_1$; and $p \wedge q$ (and therefore $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$) is exactly verified (falsified) by $s_1 \sqcup t_1$ yet merely inexactly falsified (verified) by $s_2 \sqcup t_2$.

$\{p \wedge q, \neg p \wedge \neg q\}$, however, is not a possible subset of $\mathcal{K}$ under $\mathcal{EL}_3$, and *a fortiori* $\mathcal{EL}_4$. For any two states σ and τ , if $\sigma \Vdash p \wedge q$, σ and $\tau \Vdash \neg p \wedge \neg q$, τ would have an exact verifier of $\neg p$ as part (possibly improper), which is an exact falsifier of p , and therefore an exact falsifier of $p \wedge q$. Thus τ would have an exact falsifier of $p \wedge q$ as part, i.e., is an inexact falsifier of $p \wedge q$. Therefore, any such set $\{\sigma, \tau\} \not\subseteq \mathcal{S}^{\mathcal{K}}$ under $\mathcal{EL}_3$ and $\mathcal{EL}_4$, and therefore $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$ is also impossible under those frameworks.

Likewise, we might wish find a set of formulas that is possibly a subset of $\mathcal{K}$ under $\mathcal{EL}_3$ but not under $\mathcal{EL}_4$. This is, however, impossible. In fact, any possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$, and vice versa. A set is a possible $\mathcal{K}$ -subset under a framework $\mathcal{X}_n$ if and only if there is a model M giving $\mathcal{X}_n$ such that there is a $\mathcal{S}^{\mathcal{K}} \in \Sigma_n$ where the set is a subset of $\Delta_X(\mathcal{S}^{\mathcal{K}})$. Then:

Theorem 3.1 (Equivalence of Possible $\mathcal{K}$ -subset Under $\mathcal{EL}_3$ and $\mathcal{EL}_4$). (a) Every possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$, and (b) every possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$.

Proof for (a): We prove the contrapositive. Since Γ is not a $\mathcal{K}$ -subset under $\mathcal{EL}_4$, by definition of $\Delta_{\mathcal{EL}}$, there are ϕ and ψ in Γ such that $\forall \sigma, \tau ((\sigma \Vdash \phi \ \& \ \tau \Vdash \psi) \implies \Theta_4(\sigma, \tau))$, that is, $\exists \chi ((\sigma \Vdash \chi \ \& \ \tau \Vdash \chi) \vee (\tau \Vdash \chi \ \& \ \sigma \Vdash \chi))$. For sake of simplicity, suppose s and t are the sole exact verifiers for ϕ and ψ respectively, and χ_1 is the witness of χ such that $s \Vdash \chi_1 \ \& \ t \Vdash \chi_1$. Then, exists $s' \sqsubseteq s$ and $t' \sqsubseteq t$ such that $s' \Vdash \chi_1$ and $t' \nVdash \chi_1$; which further yields $t' \nVdash \chi_1 \wedge \phi$, and therefore $t \nVdash \chi_1 \wedge \phi$, and $s' \sqcup s \Vdash \chi_1 \wedge \phi$. Now, since $s' \sqsubseteq s$, we get $s' \sqcup s = s$, and therefore $s \Vdash \chi_1 \wedge \phi$. By the definition of $\Delta_{\mathcal{EL}}$ and the supposition that s and t are unique exact verifiers of ϕ and ψ respectively, Γ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ if and only if $\neg \Theta_3(s, t)$. But since $s \Vdash \chi_1 \wedge \phi$ and $t \nVdash \chi_1 \wedge \phi$, $\Theta_{2a}(s, t)$ and therefore $\Theta_3(s, t)$. Now suppose χ_2 is the witness of χ such that $s \nVdash \chi_2$ and $t \Vdash \chi_2$. By similar reasoning, we can get the result that $t \Vdash \chi_2 \wedge \psi$ and $s \nVdash \chi_2 \wedge \psi$, thus $\Theta_{2a}(s, t)$ which implies $\Theta_3(s, t)$. In case where ϕ and ψ have multiple exact verifiers, we simply repeat the reasoning above. Therefore, any Γ such that is not a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$ is also not a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$, and since a set of formulas is either a possible $\mathcal{K}$ -subset or not, every possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$.

Proof for (b): trivial.

Since the expansion with states does not always success, the expansion with formulas must also be given a new definition:

Definition 3.8 (Operations on Belief Set with Consistency Constraints). Given a model M and a selection function γ , for any $\Sigma_n \subseteq \Sigma_1$,

$$\begin{aligned}
 (\mathcal{EL}^\oplus) \ D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) \oplus_n \phi &= \begin{cases} \gamma(D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K} \cup \{\sigma\}) \text{ s.t. } \sigma \Vdash \phi & \text{if } \mathcal{S}^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) & \text{otherwise} \end{cases} \\
 (\mathcal{EL}^\ominus) \ D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) \ominus_{\mathcal{EL}_n} \phi &= D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K} \setminus \{\sigma : \sigma \Vdash \phi\})
 \end{aligned}$$

$$(\mathcal{IL}^\oplus) \ D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K}) \oplus_n \phi = \begin{cases} \gamma(D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K} \cup \{\sigma\}) \text{ s.t. } \sigma \Vdash \phi & \text{if } \mathcal{S}^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K}) & \text{otherwise} \end{cases}$$

$$(\mathcal{IL}^\ominus) \ D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K}) \ominus_{\mathcal{IL}_n} \phi = D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K} \setminus \{\sigma : \sigma \Vdash \phi\})$$

$$(\mathcal{EL}^\otimes) \ D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) \otimes_{\mathcal{EL}_n} \phi = \gamma(D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K} * \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{IL}^\otimes) \ D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K}) \otimes_{\mathcal{IL}_n} \phi = \gamma(D_{\mathcal{IL}_n}(\mathcal{S}^\mathcal{K} * \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

And for similar reason with **Theorem 3.1**, expanding, contracting and revising with formulas on $\mathcal{K}$ under $\mathcal{EL}_3$ and $\mathcal{EL}_4$ are also equivalent. Also, it should be noticed that although the idea of Levi-identity is preserved in defining the revision with states, it does not hold in the revision with formulas. That is, not necessarily $(D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) \ominus_{\mathcal{EL}_n} \phi) \oplus_n \phi = D_{\mathcal{EL}_n}(\mathcal{S}^\mathcal{K}) \otimes_{\mathcal{EL}_n} \phi$, and likewise for $\mathcal{IL}$ -series frameworks. However, the failure of Levi-identity is required, if one wishes revision to be always successful.

3.2.1 The “Spirit” of Paraconsistency

It is shown above that the selected frameworks with locality and consistency constraints satisfy **Local Rationality**: the modelled epistemic agent’s belief set would have non-trivial subject matter; and they do not have a possible $\mathcal{K}$ -subset in the form of $\{\phi, \neg\phi\}$, and is closed under modus ponens. It should be also clear that new frameworks do equally well in satisfying the desiderata which simple frameworks could satisfy, except **Paraconsistency**.

Whether **Paraconsistency** are satisfied under $\mathcal{EL}_3, \mathcal{EL}_4$, and $\mathcal{IL}_4$ may be disputed, not because they cannot admit inconsistent belief set without explosion, but because they cannot admit logically inconsistent belief set *at all*. While the consistency constraint for $\mathcal{EL}_1$ is too weak to forbid $p \vee \neg p \in \mathcal{K}$, the consistency constraints for the remainder is too strong, that for any possible $\mathcal{K}$ -subset Γ , $\Gamma \not\vdash \perp$.

In theory, **Paraconsistency** is satisfied vacuously. Since the belief set can never be logically inconsistent, explosion also does not occur. Yet one might contend that those three theories do not capture the “spirit” of the desideratum, for the reason we employed **Paraconsistency** among our desiderata is that we wanted the modelled epistemic agent to deal with inconsistent

belief without explosion, not that we wanted to forbid the agent to have inconsistent belief at all.

A primary answer is that we need to compromise **Local Rationality** and the alleged “spirit” of **Paraconsistency**. If one value the possibility of having logically inconsistent belief highly, one could prefer $\mathcal{EL}_1$, and permit the modelled epistemic agent to believe that $\phi \wedge \neg\phi$. If one regard the rationality must be strong enough to exclude contradictory belief like $\phi \wedge \neg\phi$, or a pair of beliefs that $\phi \wedge \psi$ and that $\neg\phi \wedge \neg\psi$, consistency constraints are just too coarse-grained to permit any other kinds of logical inconsistency. It of course permits of impossible belief set, i.e., there being no possible world in the model making every belief of the modelled agent true. Yet the complaint was on not permitting the belief set to be *logically* impossible, not metaphysically.

That being said, proponents of $\mathcal{IL}$ -series might desire another kind of framework. If one holds that the belief set must be able to be logically inconsistent and that belief set need not be closed under conjunction elimination, one can embrace frameworks $\mathcal{EL}_1$. But as mentioned above, there is only one kind of frameworks among $\mathcal{IL}$ -series - $\mathcal{IL}_4$ - which is free from blatant inconsistency. If one holds that the belief set must be closed under conjunction elimination, the approach of employing consistency constraints on Σ cannot preserve the “spirit” of **Paraconsistency**, if one values.

Given such consideration, one might desire for approaches for **Local Rationality** other than consistency constraints.

3.3 Weighing and Cancellation

Yet there could be more motivations for alternative of consistency constraints. For one thing, one may find the idea that ideal epistemic agent must only have source of belief which favors only one of ϕ or $\neg\phi$ (or nothing at all) not convincing. At least we humans do not behave like that. We sometimes have reasons for both ϕ and $\neg\phi$, and after some explicit or implicit consideration, we either favor one of them, or just suspend judgment and believe neither of them. One might favor an approach with decide the belief set from a belief-state set via similar

procedures.

A related problem is that there seem no particularly good philosophical reason to think expansion with states sometimes fail. Again, at least we humans have no problem entertaining a new source of belief that contradicts with our current belief; and it is unclear why an ideal agent should be disabled in this regard.

So an immediate idea is to modify the function Δ , so it can pick at most one formula between ϕ and $\neg\phi$ when $\mathcal{S}^\mathcal{K}$ comprises verifiers for both ϕ and $\neg\phi$. But which one? Perhaps it should be what one is more *entrenched* into.

We already defined entrenchment in **Definition 2.6**. In short, given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$, the entrenchment of a formula is the number its verifiers in $\mathcal{S}^\mathcal{K}$. In a similar vein, we define the notion of *resistance* and *net entrenchment*:

Definition 3.9 (Entrenchment, Resistance, and Net Entrenchment). *Given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$,*

($E_\mathcal{E}$) *The exact entrenchment of ϕ $E_\mathcal{E}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$, and*

($E_\mathcal{I}$) *The inexact entrenchment of ϕ $E_\mathcal{I}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$, and*

($R_\mathcal{E}$) *The exact resistance of ϕ $R_\mathcal{E}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \dashVdash \phi\})$, and*

($R_\mathcal{I}$) *The inexact resistance of ϕ $R_\mathcal{I}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \dashVdash \phi\})$.*

The net entrenchment is intuitively the resistance subtracted from entrenchment. But it should be noticed that both of them may be infinite. Thus:

$$\text{The exact net entrenchment of } \phi \text{ } NE_\mathcal{E}(\phi) = \begin{cases} E_\mathcal{E}(\phi) - R_\mathcal{E}(\phi) & \text{if both are finite} \\ E_\mathcal{E}(\phi) & \text{if } E_\mathcal{E}(\phi) > R_\mathcal{E}(\phi) \text{ and } E_\mathcal{E}(\phi) \geq \aleph_0 \\ -(R_\mathcal{E}(\phi)) & \text{if } E_\mathcal{E}(\phi) < R_\mathcal{E}(\phi) \text{ and } R_\mathcal{E}(\phi) \geq \aleph_0 \\ b & \text{if } E_\mathcal{E}(\phi) = R_\mathcal{E}(\phi) \geq \aleph_0 \end{cases}$$

$$\text{The inexact net entrenchment of } \phi \text{ } NE_{\mathcal{I}}(\phi) = \begin{cases} E_{\mathcal{I}}(\phi) - R_{\mathcal{I}}(\phi) & \text{if both are finite} \\ E_{\mathcal{I}}(\phi) & \text{if } E_{\mathcal{I}}(\phi) > R_{\mathcal{I}}(\phi) \text{ and } E_{\mathcal{I}}(\phi) \geq \aleph_0 \\ -(R_{\mathcal{I}}(\phi)) & \text{if } E_{\mathcal{I}}(\phi) < R_{\mathcal{I}}(\phi) \text{ and } R_{\mathcal{I}}(\phi) \geq \aleph_0 \\ b & \text{if } E_{\mathcal{I}}(\phi) = R_{\mathcal{I}}(\phi) \geq \aleph_0 \end{cases}$$

where $\forall z \in \mathbb{Z}, b \not\preceq z$ & $b \not\preceq z$. Especially, $b \not\preceq 1$.

Metaphorically, the net entrenchment of ϕ is the quantitative representation of the "weighing" of reasons for believing ϕ or $\neg\phi$. If there are more reason to believe that ϕ than to believe that $\neg\phi$, the net entrenchment would be positive; and negative if vice versa. If there are equally much reason to believe that ϕ and to believe that $\neg\phi$, the net entrenchment would be 0 or "balanced" - and one has good reason to suspend the judgment between ϕ and $\neg\phi$.

Then the belief set $\mathcal{K}$, given a belief-state set $\mathcal{S}^{\mathcal{K}}$, would be the set of all formulas that have positive net entrenchment. This gives us two more kinds of frameworks:

Definition 3.10 (Frameworks $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$). *Given a model M ,*

The framework $\mathcal{ELW}_0$ is a quintuple $\langle M, \Sigma_0, +_0, \div, \Delta_{\mathcal{ELW}} \rangle$ where $\Delta_{\mathcal{ELW}}(X) = \{\phi : NE_{\mathcal{E}}(\phi) \geq 1 \text{ \& } \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \square(\tau \sqsubseteq \text{sm}(\phi) \text{ \& } \tau \sqsubseteq \sqcup X)\} \subseteq \mathcal{L}$.³

The framework $\mathcal{ILW}_0$ is a quintuple $\langle M, \Sigma_0, +_0, \div, \Delta_{\mathcal{ILW}} \rangle$ where $\Delta_{\mathcal{ILW}}(X) = \{\phi : NE_{\mathcal{I}}(\phi) \geq 1 \text{ \& } \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \square(\tau \sqsubseteq \text{sm}(\phi) \text{ \& } \tau \sqsubseteq \sqcup X)\} \subseteq \mathcal{L}$.

One interesting feature of $\mathcal{ELW}_0$ is that it has the same class of possible $\mathcal{K}$ -subsets with that of $\mathcal{EL}_1$.

Theorem 3.2 (Equivalence of Possible $\mathcal{K}$ -subset Under $\mathcal{ELW}_0$ and $\mathcal{EL}_1$). *(a) Every possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$ is a possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ and (b) every possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$.*

Proof for (a): We prove the stronger thesis that any belief set under $\mathcal{EL}_1$ can be a belief-set under $\mathcal{ELW}_0$. By definition, any belief set under $\mathcal{EL}_1$ $\Gamma_1 = \Delta_{\mathcal{EL}}(X)$ for some $X \in \Sigma_1$. Since

³If one wants to model a more "prudent" epistemic agent, one can adjust the clause $NE_{\mathcal{E}}(\phi) \geq 1$ by increasing the number of the right-handed side. Similarly for $\mathcal{ILW}_0$.

$\Sigma_1 \subseteq \Sigma_0$, we get $X \in \Sigma_0$; and since if there are some exact verifiers of ϕ and no exact falsifiers of ϕ in $\mathcal{S}^\mathcal{K}$ $NE_\mathcal{E}(\phi) \geq 1$, we get $\Delta_{ELW}(X) = \Gamma_1$. Thus, any belief set under $\mathcal{EL}_1$ can be a belief set under $\mathcal{ELW}_0$ in the exact same model; a fortiori, every possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$ is a possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$.

Proof for (b): Let Γ_0 be an arbitrary possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$. Given Γ_0 , we can construct a model $\mathbf{M}'$ which gives a unique exact verifier σ_ϕ for each $\phi \in \Gamma_0$ unless ϕ is a disjunction (the formal construction of such model is left for appendix). Then we check whether $\Gamma_0 \subseteq \mathcal{K} = \Delta_{\mathcal{EL}}(\mathcal{S}^\mathcal{K})$ for some $\mathcal{S}^\mathcal{K} \in \Sigma_1$. By definition of Δ_{EL} , $\Gamma_0 \subseteq \Delta_{EL}(\mathcal{S}^\mathcal{K})$ if and only if $\mathcal{S}^\mathcal{K}$ comprises σ_ϕ for all $\phi \in \Gamma_0$. Let $\mathcal{S}^{\mathcal{K}'}$ denote the smallest possible $\mathcal{S}^\mathcal{K}$ such that comprises σ_ϕ for all $\phi \in \Gamma_0$, i.e., $\mathcal{S}^{\mathcal{K}'} = \{\sigma_\phi : \phi \in \Gamma_0\}$. Thus we check whether such $\mathcal{S}^{\mathcal{K}'}$ can satisfy the following two conditions: (1) $\mathcal{S}^{\mathcal{K}'} \subseteq S$, (2) $\forall s, t \in \mathcal{S}^{\mathcal{K}'}, \neg \exists \psi (s \Vdash \psi \ \& \ t \nVdash \psi)$. Since by supposition, $\forall \phi \in \Gamma_0, \sigma_\phi \in S$, (1) can be satisfied. For (2), suppose for the contradiction that there is some χ where its exact verifier and falsifier is in $\mathcal{S}^{\mathcal{K}'}$, which makes $\{\chi, \neg\chi\} \subseteq \Delta_{\mathcal{EL}}(\mathcal{S}^{\mathcal{K}'})$. Then, by supposition, $\{\chi, \neg\chi\} \subseteq \Gamma_0$ which yields that $NE_\mathcal{E}(\chi) \geq 1$ and $NE_\mathcal{E}(\neg\chi) \geq 1$, and therefore $E_\mathcal{E}(\chi) > R_\mathcal{E}(\chi)$ and $E_\mathcal{E}(\neg\chi) > R_\mathcal{E}(\neg\chi)$. But since by definition $E_\mathcal{E}(\chi) = R_\mathcal{E}(\neg\chi)$ and $R_\mathcal{E}(\chi) = E_\mathcal{E}(\neg\chi)$, we get $E_\mathcal{E}(\chi) > R_\mathcal{E}(\chi)$ and $R_\mathcal{E}(\chi) > E_\mathcal{E}(\chi)$, contradiction. Thus (2) is also satisfied, and therefore $\mathcal{S}^{\mathcal{K}'} \in \Sigma_1$, given the canonical model $\mathbf{M}'$. Therefore, Γ_0 is also a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$, and since there was no specific supposition on Γ_0 , every possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$.

Operations with states in $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ are in many aspects similar with simple frameworks. Expansion always succeeds, and revision is *not* defined. Since there are no consistency constraints imposed on Σ_0 , it will be artificial to decide which states must be contracted before expanding with a state in order to revise with it.

Yet it is not an obvious defect, as it was in simple frameworks. In some cases, expansion alone is enough to model revision, in case where entrenchment and resistance is finite. Suppose $\mathcal{S}^\mathcal{K}$ is such that $E_\mathcal{I}(\phi) = 1$ and $R_\mathcal{I}(\phi) = 2$, and σ_1 and σ_2 are both inexact verifiers of ϕ not in $\mathcal{S}^\mathcal{K}$. Then we get $\neg\phi \in \Delta_{\mathcal{ILW}}(\mathcal{S}^\mathcal{K})$, $\neg\phi \notin \Delta_{\mathcal{ILW}}(\mathcal{S}^\mathcal{K} + \sigma_1)$ and $\phi \in \Delta_{\mathcal{ILW}}((\mathcal{S}^\mathcal{K} + \sigma_1) + \sigma_2)$. So although it may sound strange, the *expansion* postulate on expansion (see Table 1) does not

hold under $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$.

The operations with formulas are also identical with that of simple frameworks (see **Definition 2.5**). One intriguing feature of two new frameworks are that contracting with a formula that is not in $\mathcal{K}$ can still have substantial impact on $\mathcal{K}$. Also, here expanding with formulas do not always succeed if net entrenchment is small enough, contrary to the expansion with states. One might wish to modify the definition to guarantee success, for example, by defining expansion with ϕ as the decided belief set after expanding with enough number of verifiers of ϕ on $\mathcal{S}^{\mathcal{K}}$. However, this line of definitions are either insufficient to guarantee success, or leads to severe ill-definition. For one thing, the model given might just supply us with insufficient amount of verifiers compared to falsifiers. Suppose a belief-state set is given such that $E_{\mathcal{E}}(\phi) = 1$ and $R_{\mathcal{E}}(\phi) = 5$, yet the model gives only one exact verifier of ϕ in S . Then it is just no enough number of verifiers of ϕ to force the expansion to success. Therefore, as far as I can think of, the present definition is as good as it gets.⁴

Besides of operations and inconsistency, another notable feature of frameworks with weighing is that it permits the modelled agent to believe $\phi \vee \psi$ without believing either of them. In simple frameworks and frameworks with consistency constraints, the modelled agent must believe either of disjuncts to believe a disjunction, since a verifier of a disjunction is by definition a verifier of either of its disjuncts. But this may be contended, for it seems clearly possible for an ideal agent to believe a disjunction without committed to either of its disjuncts. It is particularly so when we consider $\phi \vee \neg\phi$. I do not have belief that it will rain a week later nor that it will not rain a week later, but I have the belief that either it will rain a week later or not.

$\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ neatly accounts for this kind of cases. Suppose $E_{\mathcal{E}}(\phi) = R_{\mathcal{E}}(\phi) = E_{\mathcal{E}}(\psi) = R_{\mathcal{E}}(\psi) = 1$, and $E_{\mathcal{E}}(\neg\phi \wedge \neg\psi) = 3$. Then, although both ϕ and ψ would not be included in $\mathcal{K}$ for their net entrenchment is zero, $NE_{\mathcal{E}}(\phi \vee \psi) = 1$ and therefore $\phi \vee \psi \in \mathcal{K}$. Likewise for $\mathcal{ILW}_0$. And this is also intuitively telling: even if one has equally many sources to believe ϕ and $\neg\phi$, ψ and $\neg\psi$ and therefore rejected to belief either of them, if one has less

⁴The present definition also risks being ill-defined, to be fair, if one permits the set of verifiers of a formula to be empty. But recall that the primary definition of a model employed in our paper does not permit the case (see **Definition 1.6**). If one permits empty set of verifiers, one can add a clause on the definition stating that if ϕ has no verifier, expanding with ϕ on $\mathcal{K}$ do not cause any difference.

reason to think neither ϕ nor ψ is the case than the sum of the reason to believe that ϕ and that of ψ , one should reasonably believe that either ϕ or ψ would be the case.

However, one unfortunate consequence of both frameworks is that it does not yield the closure under modus ponens of $\mathcal{K}$. It is not because it permits $\{\phi, \neg\phi \vee \psi, \neg\phi\} \subseteq \mathcal{K}$, but because it permits beliefs of disjunctions without disjuncts, as mentioned above. Suppose $NE_{\mathcal{E}}(\phi) = 1$, $NE_{\mathcal{E}}(\neg\psi) = 0$, and $E_{\mathcal{E}}(\neg\phi \wedge \psi) = 0$. Then although $NE_{\mathcal{E}}(\neg\phi \vee \psi) \geq 1$, $NE_{\mathcal{E}}(\psi) = 0$ and thus $\{\phi, \phi \vee \neg\psi\} \subseteq \mathcal{K}$ but $\psi \notin \mathcal{K}$. Likewise for $\mathcal{ILW}_0$.

Of course, one may argue that therefore the closure under modus ponens was not desirable at all. I am personally sympathetic with this view, but one may simply favor frameworks with consistency constraints.

Before moving to the conclusion, let me lay out a rough idea for final kinds of frameworks. One might think that if the modelled epistemic agent decided to believe ϕ after "weighing", the agent must "cancel out" the states in $\mathcal{S}^{\mathcal{K}}$ that falsifies ϕ , so that the cancelled states should not further affect reasonings for more complex formulas.

This is motivated by the problem that under every framework developed so far, conjunctions are too difficult to be believed. I mentioned in §1.1.2 that there may be cases where one believes ϕ and ψ but not $\phi \wedge \psi$. Nevertheless, in most cases, equipped with the belief that ϕ and ψ , the belief that $\phi \wedge \psi$ should not be too difficult to acquire. However, even the belief-state set contains exact verifiers for both ϕ and ψ that helps nothing at all on believing that $\phi \wedge \psi$. The case is even worse in frameworks with weighing, since even there are more exact verifiers for the conjuncts than their exact falsifiers, while the exact verifiers do not help for the entrenchment of the conjunction, the exact falsifiers do help for the resistance of the conjunction.

The initial idea is as follows. Given the language $\mathcal{L}$, we first sort the formulas by their depth to construct $\mathcal{L}' = \langle \Gamma^0, \Gamma^1, \Gamma^2, \dots \rangle$ where $\Gamma^n = \{\phi : \phi \in \mathcal{L} \text{ \& } \text{depth}(\phi) = n\}$. Now, given a model M and an initial belief-state set $\mathcal{S}^{\mathcal{K}}$ (for convenience, let $\mathcal{S}^{\mathcal{K}} = \mathcal{S}^{\mathcal{K}_0}$), let $\mathcal{K}_0 = \Delta_{\mathcal{ILW}}(\mathcal{S}^{\mathcal{K}_0})$ (or $\Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}})$. I will lay out the idea with the inexact notion of verification here). From $\mathcal{K}_0$ we construct a new belief-state set after cancelling all the "defeated" states in $\mathcal{S}^{\mathcal{K}}$, that is, $\mathcal{S}^{\mathcal{K}_1} = \mathcal{S}^{\mathcal{K}_0} \setminus \{\sigma : \exists \phi \in \Gamma^0 (\phi \in \mathcal{K}_0 \wedge \sigma \triangleleft \phi)\}$. With $\mathcal{S}^{\mathcal{K}_1}$ we re-decide a belief set $\mathcal{K}_1 = \Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}_1})$,

and construct $\mathcal{S}^{\mathcal{K}_2} = \mathcal{S}^{\mathcal{K}_1} \setminus \{\sigma : \exists \phi \in \Gamma^1(\phi \in \mathcal{K}_1 \wedge \sigma \triangleleft \phi)\}$, and so on. In general, for any $n > 0$, $\mathcal{K}_n = \Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}_n})$ and $\mathcal{S}^{\mathcal{K}_n} = \mathcal{S}^{\mathcal{K}_{n-1}} \setminus \{\sigma : \exists \phi \in \Gamma^n(\phi \in \mathcal{K}_{n-1} \wedge \sigma \triangleleft \phi)\}$. Under the new frameworks with cancellation, then, the belief set would be identified as $\mathcal{K}^\infty$.

But this crude idea is not satisfying. For one thing, we might wish $\mathcal{K}_0$ and $\mathcal{K}_1$ to agree with regard of atomic formulas, $\mathcal{K}_1$ and $\mathcal{K}_2$ to agree with regard to formulas which their depth is 1, and so on. Yet with such primeval mechanism one cannot guarantee the equivalence. Suppose $\Gamma^0 = \{p, q\}$, and the model gives that $|p| = \langle \{s_1, s_2\}, \{t_1\} \rangle$ and $|q| = \langle \{s_1, s_2\}, \{t_1, t_2\} \rangle$, and $\{s_1, s_2, t_1, t_2\} \subseteq \mathcal{S}^{\mathcal{K}}$. Since $NE_{\mathcal{E}}(p) = 1$, $p \in \mathcal{K}_0$ and therefore $t \notin \mathcal{S}^{\mathcal{K}_0}$. Then although $NE_{\mathcal{E}}(q) = 0$ and therefore $q \notin \mathcal{K}_0$, since $\mathcal{S}^{\mathcal{K}_1}$ comprises two exact verifiers of q and one exact falsifier of q , $q \in \mathcal{K}_1$. There are two ways this may be problematic: either because one would not wish new formulas with depth n at $\mathcal{K}_{n+1}$ which was not in $\mathcal{K}_n$; or because one might wish that if there occurs new formulas with depth n after cancelling out from $\mathcal{S}^{\mathcal{K}_n}$, the falsifiers for the new formulas must be cancelled out from future weighing.

Another trickier problem to solve is that if we define the cancellation process as above, the pleasing feature of $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ that the belief set may contain a disjunction without containing any of its disjuncts would be weaker. It would be easy to check that under those two frameworks, it was possible to believe that $\phi \vee \psi$ alongside with the belief that $\neg\phi$ and $\neg\psi$. This, however, would be impossible with the crude idea, since the falsifier of $\neg\phi$ and $\neg\psi$ would be the verifier of $\phi \vee \psi$, and therefore if some $\mathcal{K}^n$ consists of $\neg\phi$ and $\neg\psi$, it would clear all the verifiers of $\phi \vee \psi$ at the next cancellation process. This worry, of course, may be relieved by a philosophical argument that it is not desirable to allow the ideal epistemic agent to believe that $\phi \vee \psi$ alongside with the belief that $\neg\phi$ and $\neg\psi$. However, at least I do not have one.

Still, I believe that the idea of cancellation may be worth spending time developing, although it may be quite difficult to flesh out a satisfying construction.

4 Conclusion

In this paper, I developed a family of hyperintensional frameworks for belief revision by utilizing the resources of Finean truthmaker semantics. All of them (nominally, at least) satisfy **Definite Contraction**, **Logical Ignorance**, **Paraconsistency**, **Hyperintensionality**, and **Pertinent Recovery**; while **Local Rationality** is satisfied by $\mathcal{EL}_1$, $\mathcal{EL}_3$, $\mathcal{EL}_4$, $\mathcal{IL}_4$, $\mathcal{ELW}_0$, $\mathcal{ILW}_0$ with different degrees and means.

I intentionally did not develop any arguments favoring particular frameworks among them; partially because it exceeds the scope of this paper, but also because the along the truthmaker semantical approach one might be able to develop better frameworks than what is proposed in this paper.

Let me finally note one general limitation of the approach. Even in frameworks $\mathcal{ELW}_0$, $\mathcal{ILW}_0$, for the modelled epistemic agent to believe that $p \vee \neg p$ it must have non-zero entrenchment on either p or $\neg p$. But this idea may be challenged, because one might believe $p \vee \neg p$ because one knows that this is just logically true, without having any sources to believe either p or $\neg p$. And with **Definition 1.7**, it is not clear at all how to permit the mentioned case without infringing **Logical Ignorance**.

I believe that the problem should be dealt with by expanding our language to a higher-order language. Then one can model the case where one believes that $\forall \phi(\phi \vee \neg \phi)$, which every verifier of it inexactly verifies $p \vee \neg p$, but generally not p nor $\neg p$. The remaining task, then, is to develop a truthmaker semantics that can accommodate higher-order language, which can be employed to construct a framework of belief-revision with higher-order language.

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